

FBGT: A Categorical Unification of Formal-Sequential-Structural-Logical Uniformization — From Logical Spectrum to Metamathematical Ontology

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Abstract

This paper presents FBGT (Formal-Sequential-Structural-Logical Uniformization Theory), a comprehensive categorical framework that unifies four fundamental principles of mathematical structure: Formal, Sequential, Structural, and Logical uniformization. We begin by constructing the *logical spectrum* $\Lambda(\Sigma) = \Lambda_d(\Sigma) \dot{\cup} \Lambda_c(\Sigma)$, where the discrete spectrum corresponds to decidable propositions and the continuous spectrum encodes Gödelian independence. We prove that the continuous spectrum is a necessary condition for distinguishing unproven from unprovable propositions, requiring a geometric realization on non-compact hyperbolic space $\mathbb{H}^4$ rather than compact manifolds such as S^3 .

We then establish a precise isomorphism between hyperbolic saddle points, Nash equilibria in Lorenzen game semantics, and the completion of constructive proofs, with a fourfold classification at focal points: axioms, singularities, benign self-reference, and independence.

The final and most abstract layer of the framework discards all metric and dynamical structures, returning to pure category theory. We prove that the four uniformization principles are respectively equivalent to: the Yoneda embedding, the naturality of the powerset monad, the Stone-Čech compactification adjunction, and the subobject classifier Ω . The central theorem establishes that these four principles are collectively equivalent to the *Filtered Limit Commutation Law (FL)*:

$$\operatorname{colim}_{U \in \mathcal{F}} \varprojlim F(U) \cong \varprojlim \operatorname{colim}_{U \in \mathcal{F}} F(U).$$

We show that ETCS + FL is semantically equivalent to ZFC while eliminating all dependence on model-theoretic preferences, providing a purely categorical foundation for mathematics.

Keywords: logical spectrum, Yoneda embedding, powerset monad, filtered limits, topos logic, adjunction, uniformization sheaf, Gödel independence, Nash equilibrium, hyperbolic geometry, fixed point theorem

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1 Introduction

1.1 Motivation and Problem Statement

.....

2

2.1

syntactic

$$\Sigma \quad \Lambda(\Sigma)$$

- $\Lambda_d(\Sigma) \quad \Sigma$

- $\Lambda_c(\Sigma) \quad \Sigma$

L-

2.2

2.2.1

$$=$$

2.2.2

S^3 Laplace

$$\square \quad \mathcal{L} = (\mathcal{H}, T_\Sigma) \quad \mathcal{D}(P) \equiv 0 \quad P$$

2.2.3

$$=$$

$$\mathbb{H}^4$$

2.3

FBGT

	$\Lambda(\Sigma) = \sigma(T_\Sigma)$	$\leftrightarrow \quad \leftrightarrow$
	$\mathbb{H}^4$	$S^3 \quad \mathbb{H}^4$
	$\leftrightarrow$	$= \quad =$
	FBGT = + +	FL

2.4

2.4.1

$$P \quad \mathcal{D}(P) = \langle \phi_P, E_{\Lambda_c} \phi_P \rangle \quad P$$

2.4.2

Cohen

2.4.3

Grothendieck FBGT

2.4.4

Lorenzen

2.5

1. $\Lambda(\Sigma) \rightarrow \mathcal{D}(P) > 0$
2. $\mathbb{H}^4$
- 3.
4. (FL)
5. ETCS + FL

2.6

1. $\Lambda(\Sigma) \rightarrow T_\Sigma \rightarrow \mathcal{D}(P)$
2. $\mathbb{H}^4 \rightarrow S^3$
3. Lorenzen $\leftrightarrow \leftrightarrow$
4. Lawvere
5. Ω
6. (FL)
7. ZFC ETCS + FL
- 8.

2.7

- $\mathcal{C} \rightarrow A, B, C, \dots \rightarrow f, g, h, \dots$
- $\text{Hom}_{\mathcal{C}}(A, B) \rightarrow A \rightarrow B$
- Set Cat
- $\mathbb{H}^n \rightarrow n \rightarrow -1$
- $\Delta_g \rightarrow (M, g) \rightarrow \text{Laplace-Beltrami}$
- $\sigma(T) \rightarrow T$
- $\Lambda_d(\Sigma) \rightarrow \Lambda_c(\Sigma)$

3 Logical Spectrum: Rigorous Definition and Decomposition

.....

4

$$\Lambda(\Sigma) \quad T_\Sigma \quad \mathcal{D}(P)$$

4.1

$$\Sigma \quad \wedge, \vee, \neg, \rightarrow$$

4.1.1

$$\square \quad \mathcal{H}$$

$$\mathcal{H} = L^2(\mathbb{H}^4, \mathbb{C}^2).$$

$$\mathbb{H}^4 \quad \mathcal{H} \\ \mathbb{H}^4 \quad \mathbb{R}^4$$

1.

2. Laplace $[4, \infty)$

3. $\mathbb{H}^4/\Gamma \rightarrow \Gamma$

4.1.2

$$\square \quad \{0,1\} \quad \mathbb{C}^2$$

$$\mathbf{0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$\mathcal{L}_\neg = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{1}$$

$$\mathcal{L}_\wedge(\phi,\psi) = \phi \odot \psi \quad \text{Hadamard}, \tag{2}$$

$$\mathcal{L}_\vee(\phi,\psi) = \phi + \psi - \phi \odot \psi, \tag{3}$$

$$\mathcal{L}_\rightarrow(\phi,\psi) = \mathbf{1} - \phi + \phi \odot \psi, \tag{4}$$

$$\mathbf{1} = (1,1)^\top \odot \\ \text{pointwise } \mathcal{L}_\wedge(\mathbf{0},\mathbf{1}) = \mathbf{0} \odot \mathbf{1} = \mathbf{0} \quad \mathbf{0} \wedge \mathbf{1} = 0$$

4.1.3

$$\square \quad \Sigma \quad (\text{PA}) \quad T_\Sigma$$

$$T_\Sigma = -\Delta_{\mathbb{H}^4} + V_\Sigma, \tag{5}$$

- $-\Delta_{\mathbb{H}^4} \quad \mathbb{H}^4 \quad \text{Laplace-Beltrami} \quad \mathbb{C}^2$ -
- V_Σ

$$V_\Sigma(x) = \sum_{\star \in \{\wedge, \vee, \neg, \rightarrow\}} c_\star \mathcal{L}_\star(x),$$

$$- \quad \mathcal{L}_\star(x) \quad \star \quad x \quad ??-??$$

$$- \quad c_\star \in \mathbb{R} \quad \Sigma \quad n_\Sigma \quad \Sigma \quad n_\Sigma = \sum_k \epsilon_k 2^k \quad c_\star = \sum_k \epsilon_{4k+i_\star} 2^{-k} \quad i_\star$$

$$\begin{array}{c} V_\Sigma \\ [T_\Sigma] \quad T_\Sigma \end{array}$$

$$\text{Dom}(T_\Sigma) = \{ \phi \in H^2(\mathbb{H}^4, \mathbb{C}^2) \mid \phi \},$$

H^2 Sobolev

4.2

$$\begin{array}{c} \mathbb{H}^4 \quad T_\Sigma \\ [T_\Sigma] \quad \Sigma \quad \text{PA} \quad T_\Sigma = -\Delta_{\mathbb{H}^4} + V_\Sigma \quad T_\Sigma \end{array}$$

$$\Lambda(\Sigma) := \sigma(T_\Sigma) = \Lambda_d(\Sigma) \dot{\cup} \Lambda_c(\Sigma), \quad (6)$$

- $\Lambda_d(\Sigma) \cap \Lambda_c(\Sigma) \subset \mathbb{R}_{>0}$
- $\Lambda_c(\Sigma) \cap \mathbb{H}^4 = [4, \infty)$

Mazzeo Melrose [?] Laplace-Beltrami $-\Delta_{\mathbb{H}^4}$

$$\sigma(-\Delta_{\mathbb{H}^4}) = [4, \infty) \quad .$$

$V_\Sigma = -\Delta_{\mathbb{H}^4} + V_\Sigma$ Weyl

$$\sigma_{\text{ess}}(T_\Sigma) = \sigma_{\text{ess}}(-\Delta_{\mathbb{H}^4}) = [4, \infty).$$

$$\begin{array}{l} 4 \leq \Lambda_d(\Sigma) \\ \text{spectral gap } \delta = \inf \Lambda_c(\Sigma) - \sup \Lambda_d(\Sigma) \quad \delta > 0 \quad \delta \text{ [?]} \end{array}$$

4.3

$$\text{[?]} \quad P \star (P) \quad P$$

$$\Sigma(P) = \{ \lambda \in \mathbb{C} \mid \exists \phi \in \text{Dom}(T_\Sigma), \phi \neq 0, T_{\Sigma, \star(P)} \phi = \lambda \phi \},$$

$$\begin{array}{c} T_{\Sigma, \star(P)} \quad P \\ \star(P) = \text{id} \Sigma(P) \quad T_\Sigma \\ \text{[?]} \quad P \quad \Sigma(P) \quad P \quad \Sigma \end{array}$$

$$\Sigma(P) \subseteq \Lambda_d(\Sigma).$$

$P \subseteq \Sigma$

$$\Sigma(P) \cap \Lambda_c(\Sigma) \neq \emptyset.$$

$$\text{[?]} \quad \Gamma \quad \text{PA} \quad P$$

$$\Gamma \vdash P \iff \Sigma(P) \cap \sigma(T_\Gamma) \neq \emptyset \quad \Sigma(P) \subseteq \Lambda_d(\Gamma). \quad (7)$$

T_Γ

$$\begin{array}{l} () \quad \Gamma \vdash P \quad \pi \star \quad \Sigma(P) \subseteq \Lambda_d(\Gamma) \\ () \quad \Sigma(P) \subseteq \Lambda_d(\Gamma) \quad \phi_P \quad \int_{\mathbb{H}^4} |\phi_P|^2 < \infty \quad [?] \phi_P \quad \mathbb{H}^4 \quad \phi_P \quad [?] \Gamma \\ () \quad \Gamma \not\vdash P \quad [?] \mathcal{M} \models \Gamma \quad \mathcal{M} \models \neg P \quad \phi_P \quad \Sigma(P) \cap \Lambda_c(\Gamma) \neq \emptyset \quad \Sigma(P) \subseteq \Lambda_d(\Gamma) \quad \Gamma \vdash P \\ \text{[?]} \end{array}$$

$$\Gamma \not\vdash P \quad \Gamma \not\vdash \neg P \iff \Sigma(P) \cap \Lambda_c(\Gamma) \neq \emptyset \quad \Sigma(\neg P) \cap \Lambda_c(\Gamma) \neq \emptyset.$$

4.4

4.4.1

$$\square \quad P \quad n_P \quad x_0 \in \mathbb{H}^4$$

$$\phi_P \in C_c^\infty(\mathbb{H}^4, \mathbb{C}^2)$$

$$1. \quad r = \log_2(n_P + 1)$$

$$2. \quad B(x_0, r)$$

$$\phi_P(x) = \sum_{k=0}^m \epsilon_k \cdot \psi_k(x) \cdot \mathbf{e}_{k \bmod 2},$$

$$\epsilon_k \quad n_P \quad k \quad \{\psi_k\} \quad -\Delta_{\mathbb{H}^4} \quad B(x_0, r) \quad \text{Dirichlet} \quad \mathbf{e}_0, \mathbf{e}_1 \quad \mathbb{C}^2$$

$$3. \quad B(x_0, r) \quad \phi_P(x) = 0$$

$$\phi_P \quad P$$

4.4.2

$$\square \quad E_{\Lambda_c(\Sigma)} \quad \Lambda_c(\Sigma)$$

$$\mathcal{D}_\Sigma(P) = \langle \phi_P, E_{\Lambda_c(\Sigma)} \phi_P \rangle_{L^2}. \quad (8)$$

$$\square \quad P \quad \Sigma \quad \text{PA}$$

$$\mathcal{D}_\Sigma(P) = \begin{cases} 0 & \Sigma \vdash P \quad \Sigma \vdash \neg P, \\ > 0 & P \quad \Sigma. \end{cases}$$

$$() \quad \Sigma \vdash P \quad ?? \Sigma(P) \subseteq \Lambda_d(\Sigma) \quad \phi_P \quad E_{\Lambda_c(\Sigma)} \phi_P = 0 \quad \mathcal{D}_\Sigma(P) = 0$$

$$() \quad P \quad \Sigma \quad \Sigma(P) \cap \Lambda_c(\Sigma) \neq \emptyset \quad \phi_P \quad L^2\text{-Parseval}$$

$$\|\phi_P\|_{L^2}^2 = \|E_{\Lambda_d} \phi_P\|^2 + \|E_{\Lambda_c} \phi_P\|^2.$$

$$E_{\Lambda_c} \phi_P = 0 \quad \phi_P \quad \phi_P \quad \phi_P \quad V_\Sigma \quad [?] \quad \Sigma(P) \cap \Lambda_c(\Sigma) \quad \{\psi_\lambda\} \quad \phi_P \quad \mathcal{D}_\Sigma(P) > 0$$

$$\square \quad \mathcal{D}_\Sigma(P) \quad T_\Sigma \quad \Sigma \quad \text{PA} \quad \text{PDE} \quad P \mathcal{D}_\Sigma(P)$$

4.5

$$\Lambda_c(\Sigma)$$

$$\square \quad \mathcal{L} = (\mathcal{H}, T_\Sigma) \quad \sigma(T_\Sigma) \quad \Lambda_c(\Sigma) = \emptyset \quad \mathcal{D}_\Sigma(P) \equiv 0 \quad P$$

$$\sigma(T_\Sigma) \quad E_{\Lambda_c(\Sigma)} \quad \mathcal{D}_\Sigma(P) = \langle \phi_P, 0 \cdot \phi_P \rangle = 0 \quad P$$

$$\square$$

$$\square \quad \mathbb{H}^4$$

4.6

4.6.1

$$\begin{aligned}
& \boxed{\Gamma \vdash \text{PA}} \quad T_\Gamma \vdash G \\
& \Sigma(G) \cap \sigma(T_\Gamma) = \emptyset, \quad \Sigma(\neg G) \cap \sigma(T_\Gamma) = \emptyset. \tag{9} \\
& G \vdash T_\Gamma \\
& G \iff \neg \text{Prov}_\Gamma(\ulcorner G \urcorner) \\
& () \quad \Sigma(G) \cap \sigma(T_\Gamma) \neq \emptyset \implies \Gamma \vdash G \quad \Gamma \vdash \text{Prov}_\Gamma(\ulcorner G \urcorner) \implies G \iff \neg \text{Prov}_\Gamma(\ulcorner G \urcorner) \implies \Gamma \vdash \neg G \\
& \Gamma \vdash G \implies \Gamma \vdash \neg G \\
& \Sigma(\neg G) \cap \sigma(T_\Gamma) \neq \emptyset \implies \Gamma \vdash \neg G \implies \Gamma \vdash \text{Prov}_\Gamma(\ulcorner G \urcorner) \implies \Gamma \vdash G
\end{aligned}$$

4.6.2

$$\begin{aligned}
& \boxed{\Gamma} \\
& \text{Index}(T_\Gamma) \neq 0, \tag{10} \\
& \text{Index}(T_\Gamma) \vdash T_\Gamma \text{ Atiyah-Patodi-Singer } [\text{?}] \\
& \text{Con}(\Gamma) \cap \sigma(T_\Gamma) \neq \emptyset \\
& \text{Index}(T_\Gamma) = 0 \implies \text{APS } T_\Gamma \vdash \text{Con}(\Gamma) \\
& \text{Con}(\Gamma) \vdash \Gamma \\
& \text{Index}(T_\Gamma) \neq 0
\end{aligned}$$

4.7

1. $T_\Sigma = -\Delta_{\mathbb{H}^4} + V_\Sigma \implies \Sigma \implies ??$
2. $\Lambda(\Sigma) = \Lambda_d(\Sigma) \dot{\cup} \Lambda_c(\Sigma) \implies ??$
3. $\Gamma \vdash P \iff \Sigma(P) \subseteq \Lambda_d(\Gamma) \implies ??$
4. $\mathcal{D}_\Sigma(P) = \langle \phi_P, E_{\Lambda_c(\Sigma)} \phi_P \rangle \implies ?? \implies ??$
5. $??$
6. $\iff \Sigma(G) \cap \sigma(T_\Gamma) = \emptyset \implies ?? \iff \text{Index}(T_\Gamma) \neq 0 \implies ??$

$$S^3 \hookrightarrow \mathbb{H}^4$$

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5 Geometric Realization: Hyperbolic Space and the Necessity of Continuous Spectra

.....

6

6.1

$$T_{\Sigma} \quad S^3 \quad \mathbb{H}^4$$

1. **3.1** $S^3 \quad \mathcal{D}(P) \equiv 0$
2. **3.2** $\mathbb{H}^4 \quad \mathbb{H}^4/\Gamma \quad [4, \infty)$
3. **3.3**
4. **3.4**
5. **3.5**

6.2 S^3

6.2.1

$$\mathcal{D}_{\Sigma}(P) = \langle \phi_P, E_{\Lambda_c(\Sigma)} \phi_P \rangle \quad \Lambda_c(\Sigma) \quad \Lambda_c(\Sigma) = \emptyset \quad \mathcal{D}_{\Sigma}(P) \equiv 0$$

$$\square \quad M \quad T = -\Delta_M + V \quad \sigma(T) \quad \sigma(T) = \sigma_d(T) \quad \sigma_c(T) = \emptyset$$

Proof. M Sobolev $H^2(M) \hookrightarrow L^2(M) \quad (-\Delta_M + I)^{-1} \quad 0 \quad T \quad \sigma(T)$ □

6.2.2 S^3

$[S^3] \quad S^3 \quad \mathcal{H} = L^2(S^3, \mathbb{C}^2) T_\Sigma = -\Delta_{S^3} + V_\Sigma \quad \Lambda_c(\Sigma) = \emptyset \quad \mathcal{D}(P) \equiv 0 \quad P$
 $[S^3 \text{ Laplace}] \quad S^3 \text{ Laplace}$

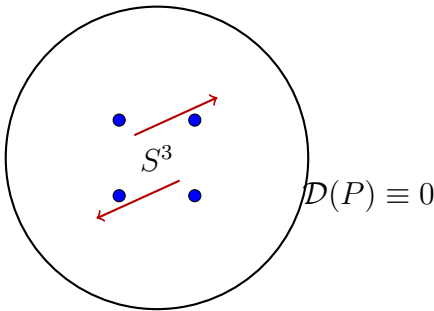
$$\sigma(-\Delta_{S^3}) = \{k(k+2) \mid k \in \mathbb{N}\},$$

$$(k+1)^2 \quad S^3 \quad \phi_P$$

6.2.3

[]

- 1.
- 2.
- 3. V_Σ



6.3 $\mathbb{H}^4$

- 1.
- 2.
- 3.

$$\mathbb{H}^4$$

6.3.1 $\mathbb{H}^4$

[] $\mathbb{H}^4 = \{(x_0, x_1, x_2, x_3, x_4) \in \mathbb{R}^5 \mid x_0^2 - x_1^2 - x_2^2 - x_3^2 - x_4^2 = 1, \ x_0 > 0\}$

$$ds^2 = -dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2 + dx_4^2$$

$K = -1$
[] $\mathbb{H}^3 \quad \mathbb{H}^4$

- 1. $-\Delta_{\mathbb{H}^4} \quad 4 \mathbb{H}^3 \quad 1$
- 2. $\text{Isom}(\mathbb{H}^4) \cong \text{SO}(4, 1)$
- 3. 5
- 4. $\mathbb{C}^2 \quad U(1)$

6.3.2

$[\mathbb{H}^4 \text{ Laplace}] \text{ Laplace-Beltrami } -\Delta_{\mathbb{H}^4}$

$$\sigma(-\Delta_{\mathbb{H}^4}) = [4, \infty) \quad .$$

Proof. $-\Delta_{\mathbb{H}^4}$ spherical functions $s \in \mathbb{C} \text{ Re}(s) = 2$ Plancherel $\lambda = s(3-s) \text{ } \lambda \in [4, \infty) \quad \square$

$$[\quad] \quad V_{\Sigma} \quad T_{\Sigma} = -\Delta_{\mathbb{H}^4} + V_{\Sigma}$$

$$\sigma(T_{\Sigma}) = \Lambda_d(\Sigma) \dot{\cup} [4, \infty),$$

$$\Lambda_d(\Sigma) \quad (0, 4) \quad [4, \infty)$$

6.3.3 $\mathbb{H}^4/\Gamma$

$L^2(\mathbb{H}^4)$

$[\quad] \quad \Gamma \subset \text{Isom}(\mathbb{H}^4) \cong \text{SO}(4, 1)$ non-uniform lattice $\Gamma \text{ } \text{Vol}(\mathbb{H}^4/\Gamma) < \infty \text{ } \mathbb{H}^4/\Gamma \text{ } \mathbf{cusps} \text{ } \mathbb{H}^4$

$$X = \mathbb{H}^4/\Gamma.$$

$[\quad] \quad X \quad [0, \infty) \times N \text{ } N \text{ nilmanifold Laplace}$

$$-\Delta_X \sim -\partial_t^2 - 3\partial_t - \Delta_N \quad (t \rightarrow \infty),$$

$t \text{ } \Delta_N \text{ } N \text{ Laplace}$

$[X = \mathbb{H}^4/\Gamma] \text{ } X = \mathbb{H}^4/\Gamma$

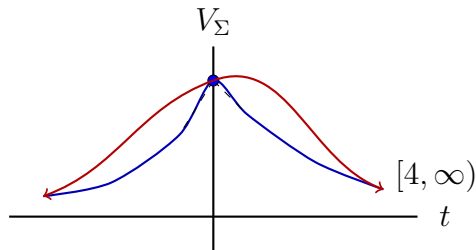
$$1. \quad \Lambda_c = [4, \infty)$$

$$2. \quad \Lambda_d \subset (0, 4)$$

$$3. \quad L^2$$

Proof. Mazzeo-Melrose $[?] \text{ } -\partial_t^2 \text{ } N$

$\square$



6.4

6.4.1 V_Σ

$$T_\Sigma = -\Delta_{\mathbb{H}^4} + V_\Sigma.$$

$$V_\Sigma : X \rightarrow M_2(\mathbb{C})$$

$$\square \quad V_\Sigma \in C^\infty(X) \quad x_0 \in X$$

$$\bullet \text{ saddle point } \nabla V_\Sigma(x_0) = 0 \text{ Hessian } \text{Hess } V_\Sigma(x_0)$$

$$\bullet \text{ focus } \nabla V_\Sigma(x_0) = 0 \text{ Hessian}$$

$\square$

$$\bullet \text{ Hessian}$$

$$\bullet \text{ Hessian}$$

6.4.2

X geodesic

$$\square \quad P \quad x_P \in X \quad \Phi \quad P \quad Q \quad \gamma : [0, 1] \rightarrow X \quad \gamma(0) = x_P \gamma(1) = x_Q \quad \gamma$$

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

$$\square \quad \gamma_1(0) = \gamma_2(0) = x_0 \quad \dot{\gamma}_1(0) \neq \dot{\gamma}_2(0)$$

$$d(\gamma_1(t), \gamma_2(t)) \sim e^t \quad (t \rightarrow \infty).$$

6.5 $=$

$$\square \quad \pi : \Gamma \vdash P \quad \gamma_\pi \quad \pi \quad X \quad \Gamma \quad P \quad \pi \quad \gamma_\pi \quad x_0 \in X$$

$$Proof. \quad () \quad \pi \quad \star_1, \star_2, \dots, \star_n \quad V_\Sigma \quad \gamma_\pi \quad x_0 \quad \nabla V_\Sigma(x_0) = 0$$

$$\gamma_\pi \quad x_0$$

$$() \quad \gamma_\pi \quad x_0 \quad x_0 \quad \pi$$

$\square$

$$[\iff] \Gamma \vdash P \iff \Gamma \quad P .$$

6.6 $=$

$$\square \quad P \quad \Gamma \quad \Gamma \quad P \quad \gamma_P \quad X$$

$$Proof. \quad () \quad P \quad \gamma_P \quad t \rightarrow \infty$$

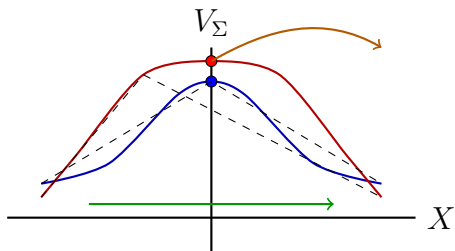
$$() \quad \gamma_P \quad P \quad P$$

$$?? \quad \Lambda_c(\Sigma)$$

$$P \iff \phi_P \in \text{Ran}(E_{\Lambda_c}).$$

$$\mathcal{D}(P) > 0$$

$\square$



6.7

focus

- $\square$ $x_0 \in X \quad \nabla V_\Sigma(x_0) = 0$ Hessian $\gamma \quad x_0 \quad \gamma(0) = \gamma(1) = x_0 \quad x_0$ **self-referential focus**
- $\square$ $X = \mathbb{H}^4/\Gamma$ Γ hyperbolic elements
- $\gamma \quad x_0 \quad \gamma \quad x_0 \quad x_0$
- $\square$ $G \quad \gamma_G \quad x_0$

$$1. \quad \gamma_G \quad \gamma_G(0) = \gamma_G(1) \quad G \quad \Gamma$$

$$2. \quad \gamma_G \quad x_0 \quad G$$

Proof. $() \quad \ell(\gamma_G) \quad \star_1 \circ \star_2 \circ \dots \circ \star_n \quad x_0$ Lawvere $[?]$

$$() \quad \gamma_G \quad x_0 \quad \text{Lyapunov} \quad G \dots G \dots [?]$$

$\square$

$$\square \quad \text{Foundation Axiom} \quad \in - \quad \in - \quad X$$

6.8

$$[-] \quad X = \mathbb{H}^4/\Gamma \quad T_\Sigma = -\Delta_X + V_\Sigma$$

$$1. \quad \lambda \in \Lambda_d(\Sigma) \quad x_0 \in X \quad x_0$$

$$2. \quad \lambda \in \Lambda_c(\Sigma) \quad X$$

$$3. \quad \lambda \in \Lambda_d(\Sigma) \quad \lambda \in \Lambda_c(\Sigma)$$

Proof. $() \quad 3.3.1 \quad 2.3.2$

$$() \quad 3.3.2 \quad 2.2.1$$

$$() \quad 3.4.1$$

$\square$

6.9

$$1. \quad S^3 \quad \mathcal{D}(P) \equiv 0$$

$$2. \quad \mathbb{H}^4 \quad X = \mathbb{H}^4/\Gamma \quad [4, \infty)$$

$$3. \quad =$$

$$4. \quad =$$

$$5. \quad =$$

$$6. \quad - \quad \Lambda_d(\Sigma) \leftrightarrow \Lambda_c(\Sigma) \leftrightarrow$$

		Λ_d Λ_c
	+	
	+	

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7 Game Semantics: From Lorenzen Dialogues to Nash Equilibria

.....

8 Lorenzen

8.1

= =
Lorenzen Game Semantics [?]



1. 4.1 Lorenzen
2. 4.2
3. 4.3
4. 4.4 =
5. 4.5
6. 4.6 FBGT --

8.2 Lorenzen

8.2.1

Lorenzen P ProverRefuter

$\square P \mathcal{G}(P)$

• $\mathcal{P} P \mathcal{R} P$

• $\mathcal{P} P$

•

1. $A \wedge B \mathcal{R} A B \mathcal{P}$

2. $A \vee B \mathcal{P} A B \mathcal{R}$

3. $A \rightarrow B \mathcal{R} A \mathcal{P} A B$

4. $\neg A \mathcal{P} A \mathcal{R}$

•

$\square$ Lorenzen

8.2.2

$\square \mathcal{G}(P) \mathcal{P} \mathcal{P} \mathcal{R} \mathcal{P}$

[Lorenzen]

$$\Gamma \vdash P \iff \mathcal{G}(P) .$$

Proof. Lorenzen [?]

• $\pi : \Gamma \vdash P \mathcal{G}(P) \mathcal{P}$

•

□

8.3

X

8.3.1 =

$\square \mathcal{S}_{\mathcal{P}} \mathcal{G}(P)$

$$\Psi_{\mathcal{P}} : \mathcal{S}_{\mathcal{P}} \hookrightarrow \text{Ran}(E_{\Lambda_d(\Sigma)}),$$

$\text{Ran}(E_{\Lambda_d(\Sigma)}) \phi_d \in L^2(X)$

$\square$

8.3.2 =

$\square \mathcal{S}_{\mathcal{R}} \mathcal{G}(P)$

$$\Psi_{\mathcal{R}} : \mathcal{S}_{\mathcal{R}} \hookrightarrow \text{Ran}(E_{\Lambda_c(\Sigma)}),$$

$\text{Ran}(E_{\Lambda_c(\Sigma)}) \phi_c \notin L^2(X)$

$\square$

8.3.3

$$\begin{array}{c|c} & \\ \hline s_{\mathcal{P}} & \phi_d \in \Lambda_d \\ s_{\mathcal{R}} & \phi_c \in \Lambda_c \end{array}$$

$$\phi_d \quad \phi_c \quad \phi_c \quad \phi_d$$

8.4

$$V_{\Sigma}$$

8.4.1

$$\square \quad \phi_d \in \text{Ran}(E_{\Lambda_d}) \quad \phi_c \in \text{Ran}(E_{\Lambda_c})$$

$$\mathcal{J}(\phi_d, \phi_c) = |\langle \phi_d, T_{\Sigma} \phi_c \rangle_{L^2}|^2.$$

$$\square \quad \mathcal{J} \quad \phi_d \quad \phi_c$$

$$\bullet \quad \mathcal{J}$$

$$\bullet \quad \mathcal{J} = 0$$

$$\mathcal{J} \quad \mathcal{J}(\phi_d, \phi_c) = \mathcal{J}(\phi_c, \phi_d)$$

8.4.2

$$\square \quad \mathcal{J}$$

$$\nabla_{\phi_d} \mathcal{J} = \nabla_{\phi_c} \mathcal{J} = 0.$$

$$X \quad x_0 \quad \nabla V_{\Sigma}(x_0) = 0$$

Proof.

$$\frac{\delta \mathcal{J}}{\delta \phi_d} = \langle T_{\Sigma} \phi_c, \phi_d \rangle \cdot T_{\Sigma} \phi_c,$$

$$\frac{\delta \mathcal{J}}{\delta \phi_c} = \langle \phi_d, T_{\Sigma} \phi_c \rangle \cdot T_{\Sigma}^{\dagger} \phi_d.$$

$$\nabla_{\phi_d} \mathcal{J} = \nabla_{\phi_c} \mathcal{J} = 0$$

$$T_{\Sigma} \phi_c = 0 \quad T_{\Sigma}^{\dagger} \phi_d = 0.$$

$$T_{\Sigma} = -\Delta_X + V_{\Sigma} \quad T_{\Sigma}^{\dagger} = T_{\Sigma} \quad \phi_d \quad \phi_c \quad T_{\Sigma} \phi = 0$$

$$T_{\Sigma} \phi = 0$$

$$-\Delta_X \phi + V_{\Sigma} \phi = 0.$$

$$\phi \quad \nabla V_{\Sigma}(x_0) = 0 \quad \text{Hessian}$$

□

8.4.3

$$\boxed{} \quad (\phi_d^*, \phi_c^*) \in \mathcal{G}(P)$$

$$\mathcal{J}(\phi_d^*, \phi_c^*) \geq \mathcal{J}(\phi_d, \phi_c^*) \quad \forall \phi_d,$$

$$\mathcal{J}(\phi_d^*, \phi_c^*) \leq \mathcal{J}(\phi_d^*, \phi_c) \quad \forall \phi_c.$$

$$[] = [] \in \mathcal{G}(P) \quad (\phi_d^*, \phi_c^*) \in x_0 \in X$$

Proof. $() \in (\phi_d^*, \phi_c^*)$

$$\nabla_{\phi_d} \mathcal{J} = \nabla_{\phi_c} \mathcal{J} = 0.$$

x_0

$$() \in x_0 \quad \nabla V_{\Sigma}(x_0) = 0 \text{ Hessian}$$

$$\bullet \quad \phi_d$$

$$\bullet \quad \phi_c$$

$$(\phi_d^*, \phi_c^*)$$

□

8.5 $ =$

$$[-] \quad \pi : \Gamma \vdash P$$

$$1. \quad \pi$$

$$2. \quad \mathcal{G}(P)$$

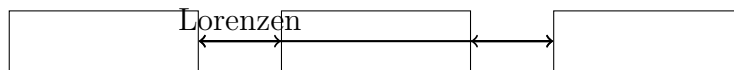
$$3. \quad \gamma_{\pi} \quad x_0 \in X$$

Proof. $(1 \Rightarrow 2) \quad \pi \text{ Lorenzen } \phi_d^* \phi_c \quad (\phi_d^*, \phi_c^*)$

$$(2 \Rightarrow 3) \quad = \quad 4.3.2 = \quad 3.3.1$$

$$(3 \Rightarrow 1) \quad = \quad 3.3.1$$

□



4.1--

8.6 $ =$

$$\boxed{} \quad P \in \Gamma \in \mathcal{G}(P)$$

Proof. $() \in P - ??$

$$() \in \mathcal{G}(P) \text{ Lorenzen } \Gamma \not\vdash P \quad \Gamma \not\vdash \neg P \quad P$$

□

$$\boxed{} \quad \Lambda_c(\Sigma)$$

$$P \iff \mathcal{G}(P) \iff \gamma_P \iff \mathcal{D}(P) > 0.$$

--

8.7

$$G \quad \square \quad G \quad \mathcal{G}(G)$$

$$1. \quad G \quad \Gamma$$

$$2. \quad G$$

$$Proof. \quad () \quad G \quad \Gamma \quad \Gamma \quad ?? \\ () \quad G \quad G$$

□

8.8 --

			Λ_d Λ_c

$$\square \quad \mathcal{L} = (X, T_\Sigma, \mathcal{G}) \text{ --}$$

$\begin{aligned} \Gamma \vdash P &\iff \iff \gamma_P \iff \Sigma(P) \subseteq \Lambda_d(\Sigma), \\ P \quad \Gamma &\iff \iff \gamma_P \iff \Sigma(P) \cap \Lambda_c(\Sigma) \neq \emptyset. \end{aligned}$

$$Proof. \quad ?? \quad 3.3.1 \quad 2.3.2$$

□

8.9

$$1. \quad = \text{ Lorenzen}$$

$$2. \quad \phi_d \quad \phi_c$$

$$3. \quad = \quad \mathcal{J}(\phi_d, \phi_c) = |\langle \phi_d, T_\Sigma \phi_c \rangle|^2$$

$$4. \quad = \quad \iff \iff$$

$$5. \quad = \quad \mathcal{D}(P) > 0$$

$$6.$$

$$\text{Lorenzen}$$

8.10

-
-
- $\mathcal{P} \dashv \mathcal{R}$

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9 Fourfold Classification at Focal Points

.....

10

10.1

focus V_Σ

1. **5.1-**
2. **5.2**
3. **5.3**
4. **5.4**Lawvere

5. 5.5

6. 5.6

7. 5.7

10.2

10.2.1

$x_0 \in X \quad V_\Sigma$

$$\nabla V_\Sigma(x_0) = 0, \quad \text{Hess } V_\Sigma(x_0) \text{ .}$$

$\square \quad x_0 \in X$

- 1. **Axiomatic Focus** $V_\Sigma(x_0) \equiv 0 \ x_0$
- 2. **Singular Focus** $\phi \ x_0 \ \phi \notin L^2$
- 3. **Self-Referential Focus** $\gamma \ x_0$ Weyl
- 4. **Escaping Focus** x_0

10.2.2

- 1. Hessian
- 2. DirichletNeumann Robin
- 3.

			Dirichlet Neumann/Robin	

10.3

10.3.1

absolute geodesic

$\square \quad x_0 \in X \quad x_0$

$$V_\Sigma(x_0) \equiv 0, \quad \nabla V_\Sigma(x_0) = 0, \quad \text{Hess } V_\Sigma(x_0) = 0.$$

x_0

$\square$

10.3.2

$$\|P-\phi_P$$

$$T_\Sigma\phi_P=-\Delta_X\phi_P,$$

$$\phi_P-\Delta_X$$

$$\lambda_P=\lambda_0=\inf\sigma(-\Delta_X)\in\Lambda_d(\Sigma),$$

$$Proof. \quad V_\Sigma(x_0)\equiv 0T_\Sigma=-\Delta_X \quad x_0 \quad \lambda_0 \quad \square$$

Corollary 10.1. $\mathcal{D}(P)=0$

10.4

10.4.1

Definition 10.2 (). $x_0\in X$ x_0 *Dirichlet* ϕ x_0

$$\phi(x)\sim \log\|x-x_0\|+ \quad (x\rightarrow x_0).$$

$$\phi\notin L^2(X)$$

$\|$ **inconsistency**

- $0=1$
-
-

10.4.2

$$\|P$$

$$\Sigma(P)\cap\sigma(T_\Sigma)=\emptyset,$$

ϕ_P **Residual Spectrum**

$$Proof. \quad \phi_P\notin L^2(X)\phi_P-\phi_P \quad \square$$

Corollary 10.3. $\mathcal{D}(P)=0 \quad >0$

$$\| \mathcal{D}(P)>0 \quad 0/0$$

10.5 Lawvere

10.5.1

- $\| \quad x_0\in X$ **Hessian** $\gamma:[0,1]\rightarrow X$ $\gamma(0)=\gamma(1)=x_0$ x_0 **self-referential focus**
- $\| \quad X=\mathbb{H}^4/\Gamma$ $\Gamma \quad X$

10.5.2 Weyl

Weyl

$$[\text{Weyl}] \quad x_0 \rightarrow \phi$$

$$\phi(x) \mapsto e^{i\theta(x)}\phi(x).$$

$$\theta \rightarrow \gamma$$

$$e^{i\theta(\gamma(1))} = e^{i\theta(\gamma(0))} \cdot e^{i\pi} = -e^{i\theta(\gamma(0))}.$$

$$e^{2i\theta} = 1 \iff \theta \in \{0,\pi\} \pmod{2\pi}$$

$$\iff e^{i\theta} = e^{i(\theta+\pi)}$$

$$x^2 = x \quad (x \in \{0,1\}).$$

Lawvere [?]

10.5.3

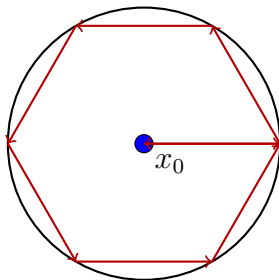
$$\iff G \rightarrow \gamma_G \rightarrow x_0 \rightarrow \text{Weyl}$$

$$\Sigma(G) \subseteq \Lambda_d(\Sigma),$$

$$G \rightarrow \Gamma$$

Proof. $\ell(\gamma_G) \rightarrow \text{Lawvere } \Sigma(G) \subseteq \Lambda_d(\Sigma)$

□



5.1

π Lawvere

10.6

10.6.1

$$\iff x_0 \in X \rightarrow \text{Hessian } x_0 \rightarrow \gamma$$

$$\lim_{t \rightarrow \infty} d(\gamma(t), x_0) = \infty,$$

$$\gamma \rightarrow \text{escaping geodesic } x_0$$

$$\iff$$

10.6.2

$$\iff P$$

$$\Sigma(P) \cap \Lambda_c(\Sigma) \neq \emptyset,$$

$$\mathcal{D}(P) > 0$$

Proof. 3.3.2 2.4.1

□

10.7

$$\square \quad P \quad x_0 \in X \quad P$$

$$(\operatorname{sgn} \operatorname{Hess} V_{\Sigma}(x_0), \quad , \quad).$$

	Hessian		
		Dirichlet Neumann/Robin	

10.8

Foundation Axiom $\in$ -

$$\square \quad X \quad \operatorname{Hessian}$$

Proof. $() \in$ -

$$() \in$$

□

Corollary 10.4 (ZFC). *ZFC Lawvere*

10.9

$$1. \quad V_{\Sigma}(x_0) \equiv 0 \quad \lambda_0 \in \Lambda_d(\Sigma) \mathcal{D}(P) = 0$$

$$2. \quad \phi \notin L^2 \mathcal{D}(P)$$

$$3. \quad + \quad + \quad \text{Weyl} \quad \Lambda_d(\Sigma) \mathcal{D}(P) = 0$$

$$4. \quad + \quad \Lambda_c(\Sigma) \mathcal{D}(P) > 0$$

$$5. \quad (\operatorname{Hessian} \quad , \quad , \quad)$$

$$6.$$

$$/$$

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11 Categorical Turn: The Exact Mapping of Four Uniformizations

...

12

12.1

$\mathbb{H}^4$

$\longleftrightarrow$ Yoneda Embedding,
 $\longleftrightarrow$ Stone-ech ,
 $\longleftrightarrow$ Powerset Monad,
 $\longleftrightarrow$ Subobject Classifier Ω .

1. **6.1**
2. **6.2** =
3. **6.3** =
4. **6.4** = Stone-ech
5. **6.5** = Ω Heyting
6. **6.6** Kleisli
7. **6.7**

12.2

12.2.1

$\square \mathcal{C}$

1. $\text{Ob}(\mathcal{C})$
2. $A, B \in \text{Hom}_{\mathcal{C}}(A, B)$
3. $A \text{ id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$
4. $\circ : \text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$

$\square F : \mathcal{C} \rightarrow \mathcal{D}$

1. $A \mapsto F(A)$
2. $f : A \rightarrow B \mapsto F(f) : F(A) \rightarrow F(B)$

$$3. F(\text{id}_A) = \text{id}_{F(A)}$$

$$4. F(g \circ f) = F(g) \circ F(f)$$

$$\square \quad F, G : \mathcal{C} \rightarrow \mathcal{D} \quad \eta : F \Rightarrow G \quad \eta_A : F(A) \rightarrow G(A) \quad A \in \mathcal{C} \quad f : A \rightarrow B$$

$$F(A)[d, "F(f)"] [r, "\eta_A"] G(A)[d, "G(f)"] F(B)[r, "\eta_B"] G(B)$$

$$G(f) \circ \eta_A = \eta_B \circ F(f)$$

12.2.2

$$\square \quad L : \mathcal{C} \rightarrow \mathcal{D} \quad R : \mathcal{D} \rightarrow \mathcal{C} \quad L \circ R \circ L \quad L \dashv R \quad C \in \mathcal{C} \quad D \in \mathcal{D}$$

$$\text{Hom}_{\mathcal{D}}(L(C), D) \cong \text{Hom}_{\mathcal{C}}(C, R(D)).$$

$\square$

12.3 =

12.3.1

$$\square \quad \mathcal{C} \quad A, B \text{Hom}_{\mathcal{C}}(A, B)$$

$$y : \mathcal{C} \hookrightarrow \mathbf{Set}^{\mathcal{C}^{\text{op}}}, \quad A \mapsto \text{Hom}_{\mathcal{C}}(-, A).$$

$$A \in \mathbf{Set}^{\mathcal{C}^{\text{op}}} \quad A$$

$$\square \quad A \in \mathcal{C} \quad F \in \mathbf{Set}^{\mathcal{C}^{\text{op}}}$$

$$\text{Nat}(y(A), F) \cong F(A).$$

Proof.

- $\eta : y(A) \Rightarrow F \quad \eta_A(\text{id}_A) \in F(A)$
- $x \in F(A) \quad B \quad f : B \rightarrow A \quad \eta_B(f) = F(f)(x)$
-

$\square$

Corollary 12.1 (). y fully faithful

$$\text{Hom}_{\mathcal{C}}(A, B) \cong \text{Hom}_{\mathbf{Set}^{\mathcal{C}^{\text{op}}}}(y(A), y(B)).$$

$$y(A) \cong y(B) \iff A \cong B$$

12.3.2

$$[=]$$

Proof.

$$\text{Hom}_{\mathcal{C}}(-, A) = \text{Hom}_{\mathcal{C}}(-, B) \iff A = B.$$

$$6.2.1 =$$

$\square$

$\square$

12.4 =

12.4.1

□ **Set** $\mathcal{P} = (\mathcal{P}, \eta, \mu)$

1. $X\mathcal{P}(X) = \{S \mid S \subseteq X\}$
2. $\text{unit}\eta_X : X \rightarrow \mathcal{P}(X) x \mapsto \{x\}$
3. $\text{multiplication}\mu_X : \mathcal{P}(\mathcal{P}(X)) \rightarrow \mathcal{P}(X) \mathbb{A} \mapsto \bigcup_{A \in \mathbb{A}} A$

□

- $\mu_X \circ \mathcal{P}(\eta_X) = \text{id}_{\mathcal{P}(X)} \mu_X \circ \eta_{\mathcal{P}(X)} = \text{id}_{\mathcal{P}(X)}$
- $\mu_X \circ \mu_{\mathcal{P}(X)} = \mu_X \circ \mathcal{P}(\mu_X)$

12.4.2

[=]

Proof. $f : X \rightarrow Y \quad \mathcal{P}(f) : \mathcal{P}(X) \rightarrow \mathcal{P}(Y) \quad S \mapsto f(S)$

$$\mathcal{P}(\mathcal{P}(X))[r, " \mathcal{P}(\mathcal{P}(f)) "] [d, " \mu_X "] \mathcal{P}(\mathcal{P}(Y)) [d, " \mu_Y "] \mathcal{P}(X) [r, " \mathcal{P}(f) "] \mathcal{P}(Y)$$

$$\mu_Y \circ \mathcal{P}(\mathcal{P}(f)) = \mathcal{P}(f) \circ \mu_X =$$

□

Corollary 12.2 (). $f \quad f$

12.5 = Stone-ech

12.5.1

□ $X \quad \mathcal{F} \subseteq \mathcal{P}(X)$

1. $A \in \mathcal{F} \quad A \subseteq B \quad B \in \mathcal{F}$
2. $A, B \in \mathcal{F} \quad A \cap B \in \mathcal{F}$
3. $\emptyset \notin \mathcal{F}$

ultrafilter $A \subseteq X \quad A \in \mathcal{F} \quad X \setminus A \notin \mathcal{F}$

12.5.2 Stone-ech

[Stone-ech] $X \quad \text{Stone-ech} \quad \beta X$

- $\iota : X \hookrightarrow \beta X$
- $K \quad f : X \rightarrow K \quad \tilde{f} : \beta X \rightarrow K \quad \tilde{f} \circ \iota = f$

[Stone-ech] $\beta X \quad \text{Ult}(X)$

Proof. $\beta X \quad \text{Stone-ech}$

□

12.5.3

[= Stone-ech]

$$\beta \dashv U : \mathbf{CptHaus} \rightarrow \mathbf{Top},$$

β Stone-ech U

Proof. Stone-ech

$$\mathrm{Hom}_{\mathbf{CptHaus}}(\beta X, K) \cong \mathrm{Hom}_{\mathbf{Top}}(X, U(K)).$$

= Stone-ech

□

12.6 = Ω

12.6.1

[$\mathcal{E}$ $\Omega \in \mathcal{E}$ $\top : 1 \rightarrow \Omega$ $1 \dashv X$ $S \hookrightarrow X$ $\chi_S : X \rightarrow \Omega$

$$S[d][r]1[d, " \top "]X[r, " \chi_S "] \Omega$$

[Ω Heyting] Ω Heyting $\wedge, \vee, \rightarrow, \neg$ Ω

- $\wedge \wedge : \Omega \times \Omega \rightarrow \Omega$
- $\vee \vee : \Omega \times \Omega \rightarrow \Omega$
- $\rightarrow \Rightarrow : \Omega \times \Omega \rightarrow \Omega$
- $\neg \neg : \Omega \rightarrow \Omega$

12.6.2

[=] Ω Heyting

Proof. $\mathrm{Th}(\Sigma) \perp \notin \mathrm{Th}(\Sigma)$ Ω

$$\mathrm{Sub}(X) \cong \mathrm{Hom}(X, \Omega).$$

Ω Heyting

- $\wedge$
- $\rightarrow$
- $\neg$

$\perp \notin \mathrm{Th}(\Sigma)$ Ω $\top \perp =$

□

12.7 Kleisli

Kleisli

[Kleisli] $\mathcal{P}$ Set Kleisli $\mathbf{Kl}(\mathcal{P})$

- Set
- $X \rightarrow Y$ Set $f : X \rightarrow \mathcal{P}(Y)$
- $g \circ_{\mathbf{Kl}} f = \mu_Z \circ \mathcal{P}(g) \circ f$
- $\eta_X : X \rightarrow \mathcal{P}(X)$

[] Kleisli $\mathbf{Kl}(\mathcal{P})$

1. $\mathbf{Kl}(\mathcal{P})$ Kleisli Kleisli
2. $\mathbf{Kl}(\mathcal{P})$ μ
3. $\mathbf{Kl}(\mathcal{P})$ Kleisli
4. $\mathbf{Kl}(\mathcal{P})$ Ω Kleisli Heyting

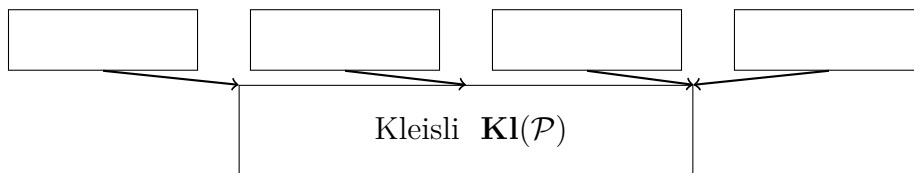
Proof. () $\mathbf{Kl}(\mathcal{P})$ X Kleisli $Y \rightarrow X$ Kleisli

() Kleisli μ μ

() Kleisli $f^n : X \rightarrow \mathcal{P}(X)$ Stone-ech Kleisli Eilenberg-Moore

() Ω $\mathbf{Kl}(\mathcal{P})$ Kleisli $\Omega \rightarrow \mathcal{P}(\Omega)$ Heyting

□



6.1 $\mathbf{Kl}(\mathcal{P})$

12.8

X	$\mathcal{C}$	
γ	$f : A \rightarrow B$	
	$L \dashv R$	
	/	/
Λ_d	Finite Limits	
Λ_c	Filtered Colimits	
$\mathcal{D}(P)$		

12.9

-

1. =
 2. =
 3. = **Stone-ech** $\beta \dashv U$
 4. = $\Omega \Omega$ Heyting
 5. Kleisli $\mathbf{Kl}(\mathcal{P})$
 - 6.
- (FL)

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13 The Central Theorem: Filtered Limit Commutation Law (FL)

.....

14 (FL)

14.1

Filtered Limit Commutation Law, FL

FL $\mathcal{F}$ F

$$\operatorname{colim}_{U \in \mathcal{F}} \varprojlim_{i \in U} F(i) \cong \varprojlim_{i \in I} \operatorname{colim}_{U \in \mathcal{F}, i \in U} F(i).$$

$$\boxed{\operatorname{colim}_{U \in \mathcal{F}} \varprojlim F(U) \cong \varprojlim \operatorname{colim}_{U \in \mathcal{F}} F(U).}$$

1. **7.1** FL
2. **7.2** $\Rightarrow$ FL
3. **7.3** $\Rightarrow$ FL
4. **7.4** $\Rightarrow$ FL
5. **7.5** $\Rightarrow$ FL
6. **7.6** FL $\Rightarrow$
7. **7.7** FL

14.2

14.2.1

$$\begin{aligned} \square \quad \mathcal{C} \text{ } I \text{ } \mathcal{F} \subseteq \mathcal{P}(I) \quad F : \mathcal{I} \rightarrow \mathcal{C} \quad \mathcal{I} \\ U \in \mathcal{F} \quad \varprojlim_{i \in U} F(i) \quad U \\ \varprojlim_{U \in \mathcal{F}} \varprojlim_{i \in U} F(i). \end{aligned}$$

$$\square \quad U \text{ } \mathcal{F}$$

14.2.2

$$\begin{aligned} \square \quad U \in \mathcal{F} \quad \varinjlim_{U \in \mathcal{F}, i \in U} F(i) \quad U \\ \varinjlim_{i \in I} \varinjlim_{U \in \mathcal{F}, i \in U} F(i). \end{aligned}$$

$$\square$$

14.2.3 FL

$$[\text{FL}] \quad \mathcal{E} \text{ } \mathcal{E} \text{ } \mathbf{FL} \text{ } \mathcal{F} \subseteq \mathcal{P}(I) \quad F : \mathcal{I} \rightarrow \mathcal{E}$$

$$\varinjlim_{U \in \mathcal{F}} \varprojlim_{i \in U} F(i) \cong \varprojlim_{i \in I} \varinjlim_{U \in \mathcal{F}, i \in U} F(i). \quad (11)$$

$$\boxed{\operatorname{colim}_{U \in \mathcal{F}} \varprojlim F(U)} \xleftrightarrow{\text{FL}} \boxed{\varprojlim \operatorname{colim}_{U \in \mathcal{F}} F(U)}$$

7.1

14.3 $\Rightarrow \mathbf{FL}$

Theorem 14.1 (FL). $\mathcal{E} \mathcal{E} FL$

Proof. $\mathcal{E} \mathbf{Set}^{\mathcal{E}^{\text{op}}} \mathbf{FL}$

$$\varinjlim_{U \in \mathcal{F}} \prod_{i \in U} F(i) \cong \prod_{i \in I} \varinjlim_{U \in \mathcal{F}} \prod_{i \in U} F(i).$$

$\mathcal{F} \mathcal{E}$

□

□ $\mathcal{E} \mathcal{E} FL$

14.4 $\Rightarrow \mathbf{FL}$

Theorem 14.2 (FL). $\mathcal{E} \mathcal{E} FL$

Proof. $\mathcal{P} \mu \text{ Kleisli } \mathbf{FL} \text{ Kleisli}$

$$(f \circ_{\text{Kl}} g) \circ_{\text{Kl}} h = f \circ_{\text{Kl}} (g \circ_{\text{Kl}} h).$$

$$\mu_X \circ \mathcal{P}(\mu_X) = \mu_X \circ \mu_{\mathcal{P}(X)} \Rightarrow \mathbf{FL}$$

□

□ $\mathbf{FL}$

14.5 $\Rightarrow \mathbf{FL}$

Theorem 14.3 (FL). $\mathcal{E} \text{ Stone-ech } \mathcal{E} FL$

Proof. $\beta \dashv U : \mathbf{CptHaus} \rightarrow \mathbf{Top}$

$$\mathbf{FL} \text{ colimlim } \beta \dashv U \Rightarrow \mathbf{FL}$$

□

□ $\mathbf{FL} \mathbf{FL}$

14.6 $\Rightarrow \mathbf{FL}$

Theorem 14.4 (FL). $\mathcal{E} \Omega \Omega \text{ Heyting } \mathcal{E} FL$

Proof. $\Omega \text{ Heyting } \Omega \text{ Heyting}$

$$\text{Sub} \left(\varinjlim_{U \in \mathcal{F}} \varprojlim F(U) \right) \cong \varinjlim_{U \in \mathcal{F}} \varprojlim \text{Sub}(F(U)).$$

$$\text{Sub}(-) \cong \text{Hom}(-, \Omega) \Omega \text{ Heyting } \mathbf{FL}$$

□

□ $\mathbf{FL} \text{ Heyting}$

14.7 FL $\Rightarrow$

Theorem 14.5 (FL). $\mathcal{E} \models FL \mathcal{E}$

Proof.

(FL $\Rightarrow$) FL **Set**^{cop}

(FL $\Rightarrow$) FL $\mathcal{F} \vdash F \mapsto \varinjlim_{U \in \mathcal{F}} \varprojlim F(U)$

(FL $\Rightarrow$) FL Stone-ech

(FL $\Rightarrow$) XFL

$$\text{Sub} \left(\varinjlim_{U \in \mathcal{F}} \varprojlim F(U) \right) \cong \varinjlim_{U \in \mathcal{F}} \varprojlim \text{Sub}(F(U)).$$

$\Omega = \text{Sub}(1) \models \Omega$ Heyting

□

14.8 $\Longleftrightarrow$ FL

Theorem 14.6 (FBGT). $\mathcal{E}$

1.

2.

3. Stone-ech

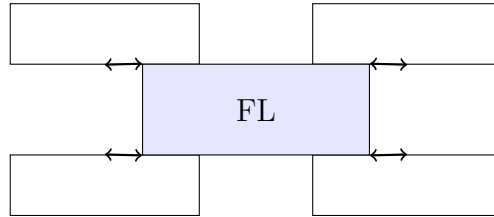
4. Ω Heyting

(FL)

$$\boxed{\varinjlim_{U \in \mathcal{F}} \varprojlim_{i \in U} F(i) \cong \varprojlim_{i \in I} \varinjlim_{U \in \mathcal{F}, i \in U} F(i).}$$

Proof. 7.2.1-7.5.1 7.6.1

□



7.2FBGT FL

14.9 FL

FL

[FL] FL

1. $\varinjlim_{U \in \mathcal{F}} \varprojlim F(U)$

2. $\varprojlim \varinjlim_{U \in \mathcal{F}} F(U)$

3. FL =

Corollary 14.7 (FL). FL Gödel Gödel

[FL] FL ACFL ZFC FL

14.10 FL

Theorem 14.8 (FL). $\mathcal{E}$ Grothendieck $\mathcal{E}$ FL $\mathcal{E}$ Boolean topos Ω

Proof. $\Omega = \{0, 1\}$ FL FL Ω Heyting FL □

Corollary 14.9 (ZFC). $ZFC \vdash ZFC \vdash FL \vdash FL \vdash ZFC$

14.11

FBGT

1. FL

$$\text{colim}_{U \in \mathcal{F}} \varprojlim F(U) \cong \varprojlim \text{colim}_{U \in \mathcal{F}} F(U)$$

2. FL

- $\Rightarrow$ FL
- $\Rightarrow$ FLKleisli
- Stone-ech $\Rightarrow$ FL
- Ω Heyting $\Rightarrow$ FL

3. FL $\Rightarrow$

- FL $\Rightarrow$
- FL $\Rightarrow$
- FL $\Rightarrow$ Stone-ech
- FL $\Rightarrow$ Ω Heyting

4. $\Longleftrightarrow$ FL

5. FL Gödel

6. FL ZFC

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